

TITLE PAGE

- Problem Statement ID –26054
- Problem Statement Title- AI-Enabled Real-Time Digital Twin System for Health Monitoring, Fault Prediction and Mission Reliability Enhancement of Aero Piston Engines used in MALE UAVs.
- Theme-Robotics and Drones
- PS Category- Software
- Team ID-
- Team Name-Innovexa



 [Demo Video](#)

 [Git Repository](#)

Problem Statement Description:

Develop an AI-enabled real-time Digital Twin for health monitoring, fault prediction, and Remaining Useful Life (RUL) estimation of aero piston engines used in MALE UAVs. The system should combine live engine telemetry, anomaly/fault detection, degradation tracking, and predictive maintenance to improve engine reliability, mission readiness, and operational safety.

Proposed Solution:

A real-time, physics-informed Digital Twin synchronizing CAN/ECU telemetry with a virtual engine state for anomaly, fault, degradation and RUL prediction. TreeSHAP + counterfactual XAI provides root-cause explanations and maintenance advisories, while the React GCS dashboard delivers real-time health, telemetry, faults, RUL and mission visualization.

Resolving the issue:

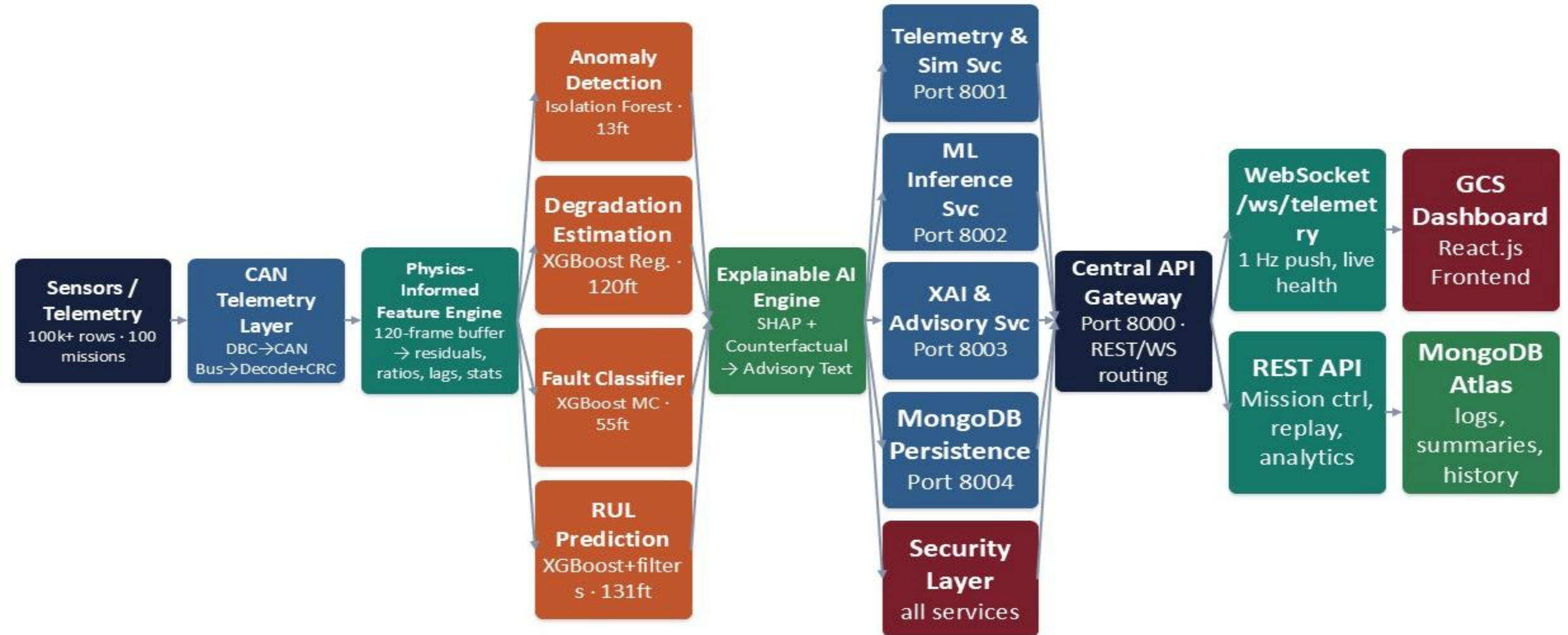
CAN Telemetry → Physics Residuals → Anomaly → Fault → Degradation → RUL → XAI → Maintenance Advisory

Enables **real-time monitoring, mission replay, fault injection, simulation, and historical health analysis** through the integrated **WebSocket + MongoDB + React GCS** architecture.

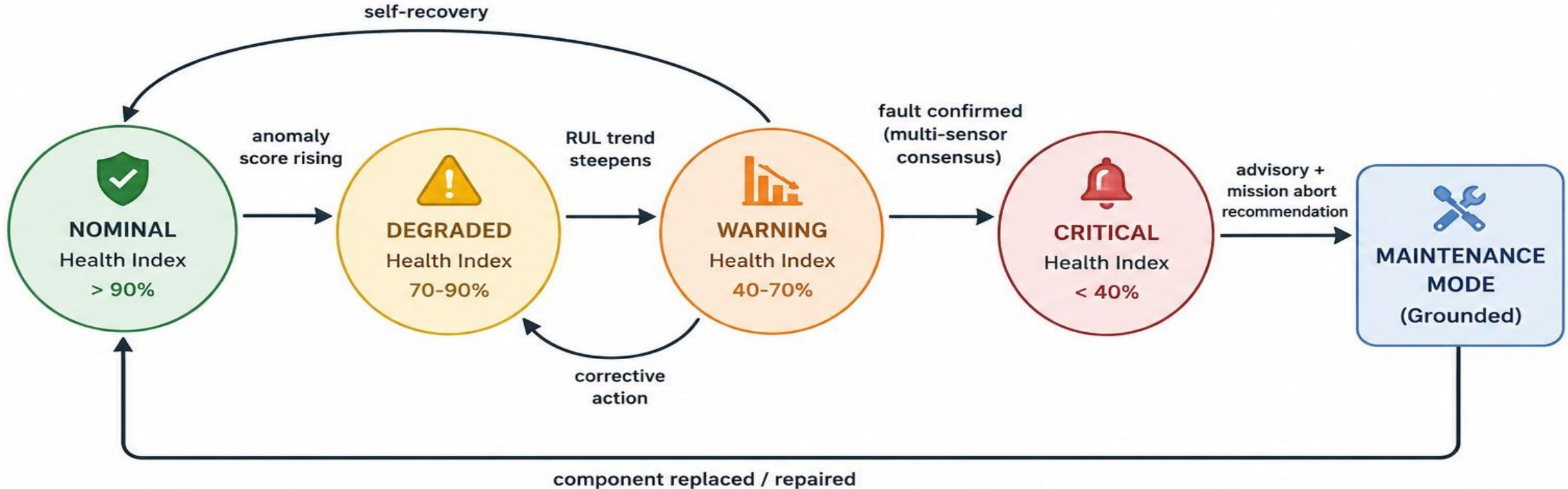
Innovation & Uniqueness

- **Physics + AI Digital Twin** — thermodynamic residuals + temporal features for context-aware engine health estimation.
- **End-to-End Prognostics** — **Anomaly → Fault → Degradation → RUL** with uncertainty bounds.
- **Explainable AI** — **TreeSHAP + Counterfactuals** for sensor-level root-cause attribution and maintenance advisories.
- **What-If & Mission Replay** — real-time fault injection, simulation and historical mission playback for diagnostic validation.
- **GCS-Ready Architecture** — **CAN 2.0B/DBC + FastAPI/WebSockets + MongoDB + React GCS**, enabling real-time telemetry, analytics and visualization.

Architecture Diagram



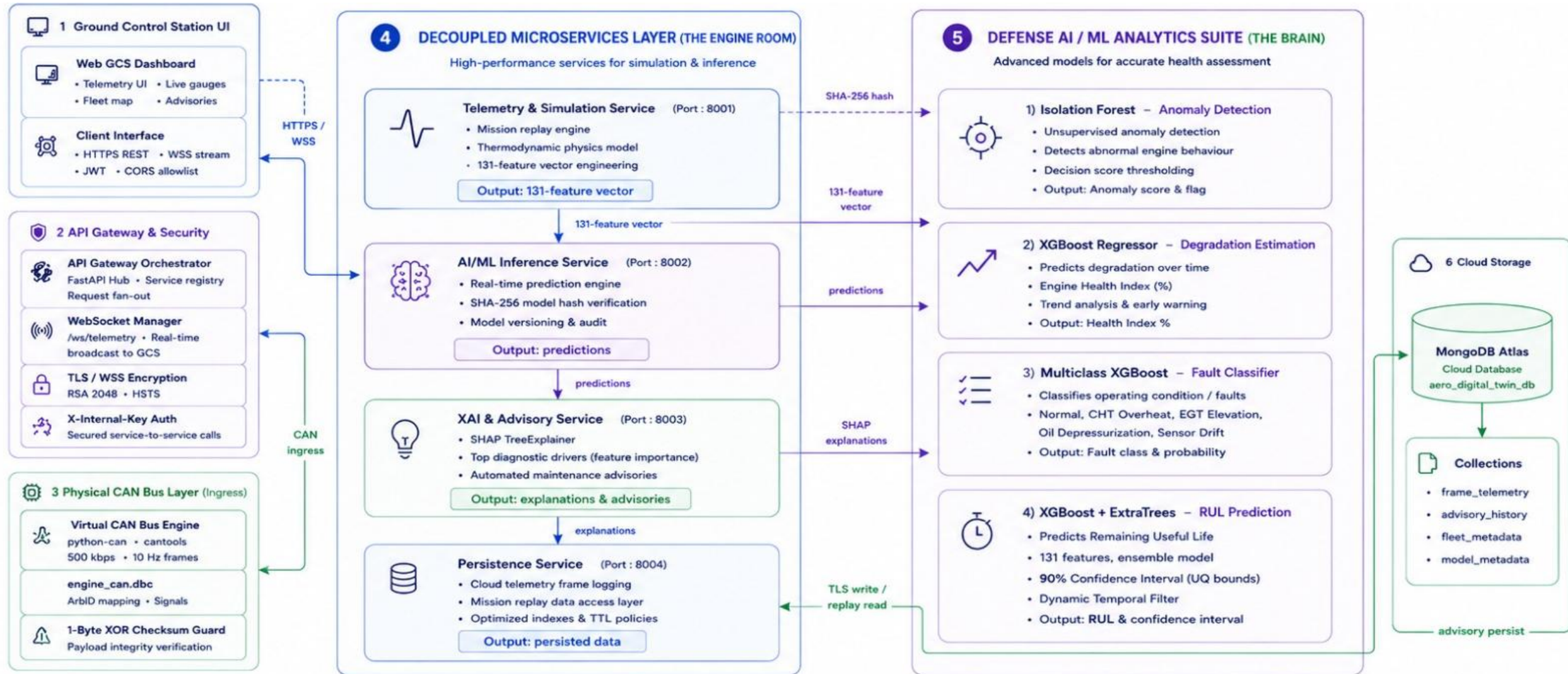
Engine Health FSM



Technology Stack



Microservices Workflow



Telemetry → Secure Microservices → AI/ML Analytics → XAI → Maintenance Advisory → Cloud Storage



Abnormal Vibration Detected

XAI Recommendations

Misfire Detected

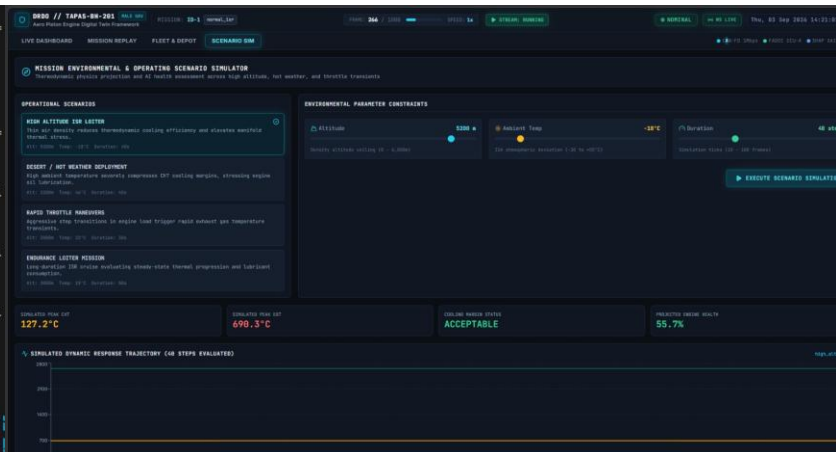
```

INFO: Unicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)
=====
ALL 5 MICROSERVICES ONLINE AND READY!
API Gateway URL: http://localhost:8000
WebSocket Stream: ws://localhost:8000/ws/telemetry
MongoDB Replay API: http://localhost:8000/api/db/mission/999/replay
Press Ctrl+C to terminate all microservices.

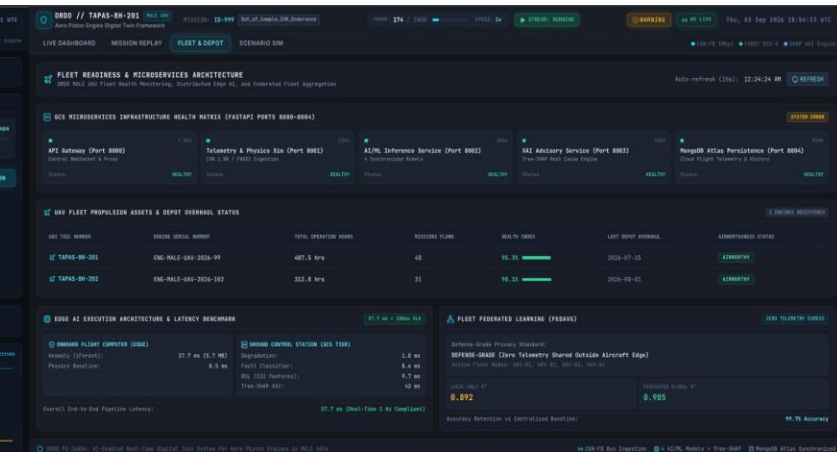
INFO: 127.0.0.1:55228 - "WebSocket /ws/telemetry" [accepted]
INFO: connection open
INFO:APIGateway:Client connected to API Gateway WebSocket Stream. Active cli
ents: 1
INFO: 127.0.0.1:50739 - "WebSocket /ws/telemetry" [accepted]
INFO: connection open
INFO:APIGateway:Client connected to API Gateway WebSocket Stream. Active cli
ents: 2
INFO: 127.0.0.1:64728 - "WebSocket /ws/telemetry" [accepted]
INFO: connection open
INFO:APIGateway:Client connected to API Gateway WebSocket Stream. Active cli
ents: 3
INFO: 127.0.0.1:60326 - "GET /missions HTTP/1.1" 200 OK
INFO: 127.0.0.1:52422 - "GET /api/missions HTTP/1.1" 200 OK
INFO: 127.0.0.1:60324 - "GET /current_frame HTTP/1.1" 200 OK
INFO: 127.0.0.1:60325 - "GET /current_frame HTTP/1.1" 200 OK
INFO: 127.0.0.1:60333 - "POST /log telemetry HTTP/1.1" 200 OK
INFO: 127.0.0.1:60332 - "POST /log telemetry HTTP/1.1" 200 OK
INFO: 127.0.0.1:60332 - "POST /log advisory HTTP/1.1" 200 OK
INFO: 127.0.0.1:60334 - "GET /missions HTTP/1.1" 200 OK

```

Microservices Checkpoints



Scenario simulator



Fleet and Depot


```
Accuracy : 0.9147
Precision: 0.8800
Recall   : 0.7873
F1-score : 0.8311
ROC-AUC  : 0.9579
PR-AUC   : 0.9169
FPR      : 0.0390
FNR      : 0.2127
```

```
              precision    recall  f1-score   support

injector_degradation      0.996      1.000      0.998       1708
lubrication_degradation    1.000      1.000      1.000       1708
      misfire              1.000      0.996      0.998        712
      normal              1.000      1.000      1.000      15401
      overheating          1.000      0.996      0.998        807
      sensor_fault         0.995      0.998      0.997        664

      accuracy                   0.999       21000
      macro avg              0.999      0.998      0.998       21000
      weighted avg           0.999      0.999      0.999       21000

Macro ROC-AUC (one-vs-rest): 1.0
```

```
| Metric | Real-time replay |
| --- | --- |
| MAE | 0.001407 |
| RMSE | 0.005255 |
| R² | 0.999626 |
| Median Absolute Error | 0.000032 |
| Maximum Absolute Error | 0.155019 |
```

```
Train
MAE: 15.384470
RMSE: 26.216987
R2: 0.987159

Validation
MAE: 34.150295
RMSE: 72.465177
R2: 0.901169

Testing
MAE: 34.095603
RMSE: 70.557896
R2: 0.900713
```

Anomaly Detection

Fault Detection

Degradation Detection

RUL

Technical Area	Dataset	Fault classification	Anomaly detection	Temporal early warning	Degradation estimation	RUL estimation	Explainability
Result Obtained	100 missions · 100,000 telemetry frames · 5 fault modes	99.9% accuracy · 0.998 macro-F1 · 1.000 macro ROC-AUC	91.47% accuracy · 83.11% F1 · 3.90% false-positive rate	5-frame confirmation logic · 3,673 confirmed anomalies on 21,000 test samples	MAE 0.00141 · RMSE 0.00525 · R² 0.99963	MAE 34.10 h · RMSE 70.56 h · R² 0.9007	Class-specific SHAP feature attribution demonstrated for fault diagnosis

Analysis of Feasibility

- Uses **existing UAV engine sensors, CAN/SocketCAN and telemetry data**, making deployment technically feasible.
- Combines **physics-based features with lightweight ML models** for real-time inference.
- Modular architecture with **FastAPI, WebSockets and MongoDB** supports scalable integration.
- Digital twin supports **real-time monitoring, simulation, mission replay and fault injection**.

Potential Challenges & Risks

- Sensor noise, drift and missing/incorrect telemetry can affect predictions.
- Limited onboard computing resources may constrain real-time AI processing.
- Insufficient real-world engine failure data can reduce model generalization.
- Cybersecurity and communication failures may affect telemetry reliability.
- False alarms or inaccurate RUL predictions could impact maintenance decisions.

Strategies to Overcome Challenges

- Apply **sensor validation, filtering and physics-based consistency checks**.
- Use **lightweight tree-based models and edge-optimized inference**.
- Combine simulation data with real flight/engine data and continuously retrain models.
- Implement **secure authenticated communication and integrity checks**.
- Use **XAI, confidence scores and uncertainty bounds** to support human decision-making.
- Validate the complete system through **mission replay and controlled fault injection**.

Impact

UAV operators: Real-time visibility of engine health, faults and mission reliability.

Maintenance teams: Early fault detection and predictive maintenance advisories.

Fleet managers: Fleet-wide health tracking and post-flight analysis.

Defence organizations: Improved mission readiness and reduced risk of engine-related mission failure.

Benefits

Safety: Detects anomalies and predicts potential failures before critical conditions.

Economic: Reduces unscheduled maintenance, downtime and component replacement costs.

Predictive Maintenance: Uses degradation trends and RUL estimation to support condition-based maintenance.

Scalability: Can be extended across multiple UAVs, missions and engine configurations.

Business Model

Target Customers: Defence organizations, UAV manufacturers, engine manufacturers, and fleet operators(DRDO).

Value Proposition: Provides real-time engine health monitoring, early fault prediction, and predictive maintenance to improve mission reliability.

Revenue Model: Annual per-UAV/fleet subscription combined with enterprise deployment and customization fees.

Go-to-Market Strategy: Start with pilot deployments and scale through defence contracts and partnerships with UAV/engine manufacturers.

Key Cost Components: AI/ML development, cloud infrastructure, edge deployment, cybersecurity, system integration, and technical support.

[1] An improved deep autoencoder framework for aviation piston engine unsupervised anomaly detection considering the real-life data characteristics
<https://www.cambridge.org/core/journals/aeronautical-journal/article/abs/an-improved-deep-autoencoder-framework-for-aviation-piston-engine-unsupervised-anomaly-detection-considering-the-reallife-data-characteristics/5CEDFA8C7CA73DC4984E00EEE760AF63>

- Proposes an improved deep autoencoder approach for detecting anomalies in aviation piston-engine data.

[2] Study of Fuel-Controlled Aircraft Engine for Fuel-Powered Unmanned Aerial Vehicle: Energy Conversion Analysis and Optimization
https://www.researchgate.net/publication/334755025_Study_of_Fuel-Controlled_Aircraft_Engine_for_Fuel-Powered_Unmanned_Aerial_Vehicle_Energy_Conversion_Analysis_and_Optimization

- Studies fuel-controlled aircraft engines for UAVs, focusing on energy conversion and performance optimization

[3] Virtual Sensors for Aero Gas Turbine Engine Health Monitoring System
https://www.drdo.gov.in/drdo/sites/default/files/technology-focus-documrnt/TF_MayJune2025.pdf

- DRDO describes an AI/ML-based comprehensive engine diagnostic and health-monitoring system using virtual sensors, operational assessment and Digital Twin technology, aimed at improving engine reliability and longevity.

[4] Fast API Documentation- <https://fastapi.tiangolo.com/>

[5] Microservice in Python using FastAPI - GeeksforGeeks
<https://www.geeksforgeeks.org/python/microservice-in-python-using-fastapi/>



[Git Repository](#)



[Demo Video](#)



[Technical Documentation](#)



[Deployment Documentation](#)